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The Frequency-Dependent Electromagnetics of the Human Body: A Comprehensive Biophysical, Dosimetric, and Regulatory Analysis

1. Introduction: The Human Organism as a Dielectric Medium

The interaction between electromagnetic fields (EMF) and the human body represents one of the most intricate frontiers in applied physics and biomedical engineering. This complexity arises not merely from the external field parameters—frequency, modulation, and intensity—but from the profound heterogeneity of the biological target itself. The human body is neither a perfect conductor nor a simple dielectric; it is a composite volume conductor characterized by a rich tapestry of tissues, each possessing unique, frequency-dependent electrical properties governed by their water content, ionic concentration, and structural organization.¹

As modern civilization advances into an era of ubiquitous wireless connectivity—spanning from the 50 Hz/60 Hz fields of power distribution grids to the sub-terahertz frequencies of 6G telecommunications—the necessity for a rigorous, exhaustive understanding of these interactions has never been more acute. This report serves as a foundational technical document, synthesizing the seminal dielectric spectroscopy data established by Gabriel et al., the biophysical mechanisms of interaction ranging from thermal thermodynamics to quantum-level ion channel dynamics, and the regulatory frameworks that attempt to bound these exposures within safety margins.³

The narrative that follows traces the journey of an electromagnetic wave as it impinges upon, penetrates, and dissipates within human tissue. It explores the "gold standard" parametric models used to simulate these events, the controversy surrounding non-thermal biological effects, and the precise dosimetric methodologies—such as the Specific Anthropomorphic Mannequin (SAM)—employed to ensure compliance. This is a study of the human body not just as a biological entity, but as a complex electromagnetic object defined by its permittivity, conductivity, and geometry.¹

2. Theoretical Foundations of Bio-Electromagnetics

To rigorously quantify the interaction of EMF with biological systems, one must first ground the analysis in the fundamental laws of classical electromagnetism. The behavior of electric and magnetic fields within biological media is governed by Maxwell's equations, which describe how these fields propagate, attenuate, and induce currents in lossy materials.

2.1 Maxwell's Equations in Biological Media

In the macroscopic limit, relevant to the frequencies discussed (10 Hz to 100 GHz), biological tissues can be treated as continuous media. Maxwell's equations in differential form provide the starting point for this description:

$$\nabla \cdot \mathbf{D} = \rho_{\text{free}}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J}_{\text{free}} + \frac{\partial \mathbf{D}}{\partial t}$$

Here, $\mathbf{E}$ and $\mathbf{H}$ are the electric and magnetic field vectors, $\mathbf{D}$ and $\mathbf{B}$ are the electric and magnetic flux densities, ρ_{free} is the free charge density, and $\mathbf{J}_{\text{free}}$ is the free current density.⁷

The specific response of biological tissue is encapsulated in the constitutive relations. Biological media are linear (at typical environmental field strengths), isotropic (largely, though muscle exhibits anisotropy), and dispersive. The relationships are:

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\mathbf{B} = \mu \mathbf{H}$$

$$\mathbf{J} = \sigma \mathbf{E}$$

In these equations, ϵ is the permittivity, μ is the permeability (which for biological tissues is approximately equal to that of free space, μ_0 , as tissues are non-magnetic), and σ is the electrical conductivity.⁹

2.2 The Wave Equation and Complex Permittivity

The interaction dynamics are most clearly seen when Maxwell's equations are combined into the vector Helmholtz equation for a time-harmonic field (assuming a sinusoidal time dependence $e^{j\omega t}$). By taking the curl of the Faraday's law equation and

substituting the Ampère-Maxwell law, we derive:

$$\nabla \times (\nabla \times \mathbf{E}) = -j\omega \mu (\sigma \mathbf{E} + j\omega \epsilon \mathbf{E})$$

Using the vector identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ and assuming a charge-free region ($\nabla \cdot \mathbf{E} = 0$), we arrive at the wave equation for a lossy medium:

$$\nabla^2 \mathbf{E} + \gamma^2 \mathbf{E} = 0$$

The propagation constant γ is defined as:

$$\gamma^2 = \omega^2 \mu \epsilon^*$$

This derivation necessitates the introduction of the **complex permittivity** (ϵ^*), a critical parameter in bio-electromagnetics that unifies the dielectric polarization and the conductive losses of the tissue. It is defined as:

$$\epsilon^*(\omega) = \epsilon'(\omega) - j\epsilon''(\omega)$$

Where:

- **$\epsilon'(\omega)$ (Real Permittivity):** Represents the ability of the tissue to store electrical energy. This is often referred to as the dielectric constant. In biological tissues, this value is extremely high at low frequencies due to interfacial polarization and drops precipitously as frequency increases.⁶
- **$\epsilon''(\omega)$ (Loss Factor):** Represents the energy dissipation within the material. This term accounts for both the dielectric relaxation losses (friction from rotating dipoles) and the static ionic conductivity.⁹

The relationship between the measured conductivity σ_{total} and the imaginary part of the permittivity is given by:

$$\sigma_{\text{total}}(\omega) = \sigma_{\text{static}} + \omega \epsilon_0 \epsilon''_{\text{relaxation}}(\omega)$$

$$\epsilon''(\omega) = \frac{\sigma_{\text{total}}(\omega)}{\omega \epsilon_0}$$

This equation reveals a fundamental truth of tissue interaction: conductivity is not a static constant. As frequency (ω) increases, the contribution of dielectric relaxation to the total conductivity grows, eventually dominating the static ionic conductivity at microwave frequencies.⁶

2.3 Power Dissipation and the Loss Tangent

The efficiency with which a biological tissue absorbs electromagnetic energy is quantified by the **loss tangent** ($\tan \delta$). This dimensionless parameter represents the ratio of the lossy reaction to the elastic (storage) reaction of the medium to the applied field:

$$\tan \delta = \frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon'}$$

A high loss tangent indicates that the material is highly absorbing (lossy), causing the wave to attenuate rapidly. For example, muscle tissue, with its high water and ion content, has a much higher loss tangent than adipose (fat) tissue or bone, meaning it absorbs energy more efficiently and acts as a shield for deeper tissues.⁹

The depth of penetration, or **skin depth** (δ), is inversely related to the loss tangent and frequency. It is defined as the distance the wave travels before its amplitude decays to $1/e$ (approximately 37%) of its surface value:

$$\delta = \frac{1}{\omega} \left[\frac{\mu \epsilon'}{2} \left(\sqrt{1 + (\tan \delta)^2} - 1 \right) \right]^{-1/2}$$

This mathematical reality dictates the biological targets of different technologies:

- **Power Lines (50 Hz):** δ is thousands of meters. The field effectively permeates the entire body volume with little attenuation.
- **Microwaves (2.4 GHz):** δ is approximately 2 cm in muscle. Heating is localized to the surface layers.
- **Millimeter Waves (60 GHz):** δ is less than 0.5 mm. Interaction is confined strictly to the stratum corneum and the cornea of the eye.¹

3. Dielectric Spectroscopy: The Gabriel Data and Dispersion Mechanisms

The theoretical framework described above relies on accurate numerical values for ϵ' and σ . The most authoritative and exhaustive characterization of these properties was conducted by C. Gabriel, S. Gabriel, and colleagues in 1996. Their work, often referred to simply as "The Gabriel Data," remains the bedrock of modern dosimetry.³

3.1 The Gabriel Methodology and Database

The Gabriel study was a monumental effort to bridge the gaps in the literature of the preceding 50 years. They employed a multi-instrument approach to cover the immense frequency range from 10 Hz to 20 GHz:

1. **Low Frequency (10 Hz – 10 MHz):** Impedance analyzers were used. A critical challenge

in this band is **electrode polarization**—the accumulation of ions at the metal-tissue interface—which can artificially inflate permittivity readings. Gabriel et al. applied rigorous correction algorithms to subtract this parasitic impedance.¹⁴

2. **Radiofrequency (10 MHz – GHz):** Open-ended coaxial probes were pressed against the tissue samples. The complex reflection coefficient was measured and converted to dielectric properties.
3. **Microwave (Up to 20 GHz):** Network analyzers with waveguide transmission lines provided high-frequency data.¹⁵

The resulting database covers over 40 distinct tissue types, from the high-water-content tissues (Cerebrospinal Fluid, Blood, Muscle) to low-water-content tissues (Bone, Fat, Tooth).

3.2 The Four Dispersions: Alpha, Beta, Delta, Gamma

The frequency dependence of biological tissues is characterized by distinct relaxation regions, known as dispersions. These occur because different polarization mechanisms have different time constants (τ)—they can only "keep up" with the oscillating field up to a certain frequency.¹⁷

3.2.1 Alpha (α) Dispersion (Hz – kHz)

Occurring at the lowest frequencies, the α -dispersion is characterized by colossal permittivity values (up to 10^6 or 10^7).

- **Mechanism:** This is primarily due to the diffusion of **counter-ions** along the surface of the cell membrane. The negatively charged cell surface attracts a cloud of positive ions (the electrical double layer). Under a slow oscillating field, this cloud distorts, creating a massive induced dipole moment.
- **Relevance:** This region dominates the interaction with power lines and is complicated by the contributions of intracellular structures like the sarcoplasmic reticulum in muscle.¹⁷

3.2.2 Beta (β) Dispersion (kHz – 100 MHz)

The β -dispersion is perhaps the most structurally significant.

- **Mechanism:** This is the classic **Maxwell-Wagner interfacial polarization**. Biological tissue is essentially a suspension of conductive spheres (cells) surrounded by insulating membranes in a conductive medium (extracellular fluid). Charges build up at the membrane interfaces, acting like a capacitor. As frequency increases into the MHz range, the reactance of the membrane capacitor ($X_c = 1/j\omega C$) decreases, effectively shorting out the membrane. The current then shifts from flowing *around* the cells to flowing *through* them.
- **Relevance:** This dispersion explains why low-frequency currents (used in TENS units) flow through the extracellular space, while RF currents (used in thermal ablation) penetrate the cell interior.¹⁷

3.2.3 Delta (Δ) Dispersion (100 MHz – 3 GHz)

Often viewed as a transition zone or a "tail" of the β -dispersion.

- **Mechanism:** It arises from the partial rotation of **macromolecular side chains** (charged groups on proteins) and the relaxation of **bound water**. Water molecules bound to proteins are rotationally hindered compared to bulk water, meaning they relax at lower frequencies (hundreds of MHz) than free water.¹⁴

3.2.4 Gamma (γ) Dispersion (> 3 GHz)

Dominating the microwave spectrum.

- **Mechanism:** This is the **dipolar rotation of free water**. Water is a polar molecule; it attempts to align with the field. At GHz frequencies, the friction from this rotation generates significant heat. The relaxation frequency of pure water is approx. 20 GHz at room temperature; in tissue, it is slightly broadened due to solutes.
- **Relevance:** This is the primary mechanism for microwave heating (ovens) and energy absorption from 5G/Wi-Fi devices.¹⁸

3.3 The 4-Cole-Cole Parametric Model

To make this complex data usable for numerical simulations (FDTD/FEM), Gabriel et al. fitted the experimental data to a summation of four Cole-Cole relaxation terms. This parametric model allows the calculation of $\hat{\epsilon}(\omega)$ at any frequency:

$$\hat{\epsilon}(\omega) = \epsilon_{\infty} + \sum_{n=1}^4 \frac{\Delta \epsilon_n}{1 + (j\omega\tau_n)^{1-\alpha_n}} + \frac{\sigma_i}{\omega\epsilon_0}$$

- ϵ_{∞} : The high-frequency limit of permittivity.
- $\Delta \epsilon_n$: The magnitude of the step in permittivity for dispersion region n .
- τ_n : The characteristic relaxation time.
- α_n : The distribution parameter (broadening) accounting for the heterogeneity of the tissue ($0 < \alpha_n < 1$).
- σ_i : The static ionic conductivity.¹⁶

Table 1: Representative Dielectric Parameters (Cole-Cole Fit)

Note: Values are illustrative of the high contrast between tissues.

Tissue	Frequency	Relative Permittivity (ϵ')	Conductivity (σ)	Mechanism Dominance
Muscle	100 Hz	> 100,000	0.2	Alpha

				(Counter-ion)
Muscle	100 MHz	70	0.7	Beta (Membrane charging)
Muscle	2.45 GHz	53	1.7	Gamma (Water rotation)
Fat	100 MHz	5	0.04	Low water content
Fat	2.45 GHz	5.3	0.1	Low water content

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4. Dosimetry: Quantifying the Energy Absorption

Dosimetry is the translation of external field exposure into internal physical quantities. The metric of choice is strictly frequency-dependent.

4.1 Specific Absorption Rate (SAR)

For frequencies from 100 kHz to 6 GHz (the bulk of radio and cellular communications), the **Specific Absorption Rate (SAR)** is the governing metric. SAR quantifies the power absorbed per unit mass of tissue (W/kg).

$$SAR = \frac{\sigma |\mathbf{E}_{int}|^2}{\rho}$$
Where $\mathbf{E}_{int}$ is the internal electric field vector RMS value. Crucially, SAR is linearly proportional to the tissue conductivity (σ) and the square of the electric field. This relationship explains why high-water-content tissues (brain, eye fluid) absorb significantly more energy than bone or fat.²⁴

Thermodynamically, SAR is related to the rate of temperature rise (dT/dt) in the absence of heat diffusion:

$$SAR = c_{heat} \frac{dT}{dt}$$
This equation is the foundation of thermal safety limits. By limiting SAR, regulators aim to limit

the temperature rise in tissues to safe levels (generally preventing a rise of $> 1^{\circ}\text{C}$).²⁶

4.2 The Transition to Power Density ($> 6\text{ GHz}$)

As frequencies rise above 6 GHz (into 5G FR2 and future 6G bands), the validity of SAR as a volumetric metric collapses. The skin depth becomes so shallow that the energy is deposited almost entirely in the cutaneous layers (skin). The concept of "mass-averaged" absorption becomes geometrically ill-defined when the interaction volume is a thin sheet.

Consequently, modern guidelines (ICNIRP 2020) shift to **Absorbed Power Density** (S_{ab}) or **Incident Power Density** (S_{inc}), measured in W/m^2 .

$$S_{\text{inc}} = \frac{|\mathbf{E}_{\text{inc}}|^2}{Z_0}$$

Where Z_0 is the impedance of free space ($377\ \Omega$). This shift acknowledges that the primary hazard at these frequencies is surface heating (corneal burns, skin burns) rather than deep tissue heating.⁵

5. Measurement Standards: The SAM Phantom and Tissue Liquids

To enforce these dosimetric limits, the industry utilizes standardized measurement protocols, most notably for certifying mobile devices (FCC/CE compliance). This involves the use of the **Specific Anthropomorphic Mannequin (SAM)**.

5.1 The Specific Anthropomorphic Mannequin (SAM)

The SAM is a physical shell phantom representing the human head. Its geometry is based on the 90th percentile dimensions of adult male heads derived from anthropometric data. This large size is a deliberate choice for conservatism: a larger head typically results in the ear (and thus the phone) being closer to the cheek, minimizing the separation distance and maximizing the measured SAR.³⁰

Critique of SAM:

While SAM is the regulatory standard, it is a homogeneous shell. It does not contain a skull, cerebrospinal fluid (CSF), or distinct grey/white matter. The phantom is filled with a single tissue-simulating liquid. Studies utilizing anatomical voxel models (like the "Virtual Population") suggest that while SAM generally overestimates SAR in the deeper brain (conservative), it may underestimate peak localized SAR in the pinna (ear cartilage) or in children, whose thinner skulls allow for deeper penetration.³²

5.2 Tissue Simulating Liquids (TSL)

The liquid filling the SAM must mimic the dielectric properties of the "average" head tissue at the specific test frequency. The formulation of these liquids is a precise chemical science,

balancing permittivity (ϵ) and conductivity (σ).

Table 2: Chemical Composition of Tissue Simulating Liquids

Frequency Band	Base Component	Additives	Function of Components
< 1 GHz	Water + Sugar (Sucrose)	NaCl, Cellulose (HEC), Bactericide	Sugar: Lowers permittivity of pure water. NaCl: Increases conductivity. HEC: Viscosity/Suspension.
1 - 2.5 GHz	Water + DGBE	NaCl (minimal)	DGBE (Diethylene-Glycol-Monobutyl-Ether): Adjusts permittivity without the bacterial issues of sugar.
> 2.5 GHz	Water + Oil Emulsion	POEM (Emulsifiers)	Oil: Needed to lower permittivity significantly. POEM: Stabilizes the oil-water mix.

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Maintenance: The water in these liquids evaporates over time, causing the permittivity to drop and conductivity to drift. Regular calibration using a dielectric probe kit is mandatory. If the liquid drifts out of tolerance (> 5% deviation), deionized water must be added and the mixture re-homogenized.³⁴

6. Biological Mechanisms: The Thermal vs. Non-Thermal Debate

The most contentious aspect of bio-electromagnetics lies in the mechanism of action. While the physics of energy absorption (dosimetry) is settled science, the biological response to

that energy remains a subject of intense scientific bifurcation.

6.1 Established Mechanism: Dielectric Heating

The consensus among major health agencies (WHO, ICNIRP, IEEE) is that the only established mechanism of harm from non-ionizing radiation is **thermal**.

- **Mechanism:** The energy dissipated by the friction of rotating water dipoles (Gamma dispersion) and oscillating ions (Alpha/Beta dispersion) is converted into heat.
- **Thresholds:** Adverse effects (e.g., behavior disruption in animals, cataracts) are observed only when tissue temperature rises by more than 1°C. This typically requires SAR values well above 4 W/kg for whole-body exposure.
- **Microwave Hearing (Frey Effect):** A unique thermal phenomenon where short, high-power pulses cause rapid thermoelastic expansion of brain tissue. This generates a pressure wave (acoustic wave) that travels to the cochlea and is perceived as a click or buzz. While "auditory," the root cause is thermal expansion, not direct nerve stimulation.³⁶

6.2 The Frontier: Non-Thermal Mechanisms

Despite the thermal consensus, a vast body of literature reports effects at levels far below the thermal threshold. These "non-thermal" effects challenge the current regulatory paradigm.

6.2.1 Ion Forced-Oscillation (IFO) & VGIC Dysfunction

Proposed by researchers like Panagopoulos, the **Ion Forced-Oscillation (IFO)** theory suggests a direct dynamical interaction between the EMF and mobile ions regulating cellular homeostasis.

- **The Physics:** Voltage-Gated Ion Channels (VGICs) in cell membranes are controlled by voltage sensors—charged helices that move in response to the transmembrane potential. The theory posits that an external oscillating field (especially ELF or ELF-modulated RF) exerts a forced oscillation on the ions (K⁺, Ca²⁺, Na⁺) passing through the channel.
- **The Critical Condition:** If the external electromagnetic force ($F = qE$) on the ions is comparable to the forces exerted by the physiological gradient, the ions can vibrate irregularly, mechanically interfering with the channel's gating mechanism.
- **Outcome:** This "irregular gating" leads to a disruption of the electrochemical balance—specifically, a pathological influx of Calcium (Ca²⁺). Intracellular calcium is a potent signaling molecule; its dysregulation is a known precursor to apoptosis and oxidative stress.³⁷

6.2.2 Reactive Oxygen Species (ROS) and DNA Damage

The "ROS Theory" serves as the biochemical bridge between EMF exposure and cellular damage.

- **The Pathway:** The dysfunction of VGICs (particularly Calcium channels) leads to cytosolic calcium overload. This activates enzymes such as **NADPH oxidase**, which

resides in the cell membrane.

- **The Oxidative Burst:** Activated NADPH oxidase produces superoxide anions ($\text{O}_2^{\bullet-}$). These free radicals initiate a cascade of oxidative stress, overwhelming the cell's antioxidant defenses.
- **Genotoxicity:** Although RF radiation is non-ionizing (it lacks the 10-12 eV required to break DNA bonds directly), ROS are chemically aggressive. They can attack the DNA backbone, causing single-strand and double-strand breaks. This provides a plausible indirect mechanism for the genotoxic effects reported in some studies (e.g., the NTP study), bypassing the "insufficient energy" argument of the thermalist view.⁴

6.2.3 Demodulation and the "Real" Signal

A critical insight in this debate is the role of modulation. Pure continuous wave (CW) RF is rare. Cellular signals (GSM, LTE, 5G) are pulsed and modulated at ELF frequencies (e.g., the 217 Hz frame rate of GSM).

Some hypotheses suggest that biological membranes, acting as non-linear rectifiers, can demodulate the signal, extracting the low-frequency envelope. This would mean the tissue is effectively exposed to ELF currents—known to be bioactive—delivered via an RF carrier. This explains why pulsed radiation is often found to be more bioactive than continuous radiation of the same average power.⁴

7. Regulatory Frameworks: ICNIRP Guidelines (1998 vs. 2020)

The regulatory landscape has evolved to reflect these physical and biological insights, though it remains conservative regarding non-thermal effects. The **International Commission on Non-Ionizing Radiation Protection (ICNIRP)** guidelines are the global benchmark.

7.1 The Evolution of Safety Limits

The ICNIRP guidelines have undergone significant revision from 1998 to 2020, driven by the need to accommodate 5G and higher frequencies.

- **ICNIRP 1998:** Focused heavily on SAR. The transition to power density was set at 10 GHz.
- **ICNIRP 2020:** Lowered the transition frequency to **6 GHz**. This was a physics-based decision: above 6 GHz, the volumetric heating model of SAR fails due to the shallow skin depth. The new guidelines rely on Incident Power Density (S_{inc}) and Absorbed Power Density (S_{ab}).⁵

7.2 Key Provisions of ICNIRP 2020

The 2020 guidelines introduce nuanced restrictions based on exposure duration and spatial averaging, acknowledging the heat diffusion capabilities of the body.

1. **Spatial Averaging (The 10g Cube):**
For local SAR, ICNIRP requires averaging over a 10g contiguous cubic mass of tissue. This

is distinct from the FCC (USA) standard, which uses a 1g volume. The 10g average is generally more lenient, as it smoothes out "hot spots" over a larger volume, based on the rationale that heat diffuses rapidly in living tissue.⁵

- 2. **Time Averaging (6 minutes vs. 30 minutes):**
 - **Local Exposure:** Averaged over 6 minutes. This corresponds to the thermal time constant of small tissue masses; it takes roughly 6 minutes for a local region to reach thermal equilibrium.
 - **Whole Body:** Averaged over 30 minutes. It takes longer for the core body temperature to rise significantly.⁵
- 3. **Brief Exposure Limits:**

For exposures shorter than 6 minutes, ICNIRP 2020 introduces a dynamic limit formula. This prevents high-intensity bursts that, while low in average power, could cause instantaneous thermal damage (e.g., the microwave hearing effect or corneal burns). The limit is proportional to $t^{0.5}$, modeling the physics of heat diffusion.⁵

Table 3: ICNIRP 2020 Basic Restrictions (General Public)

Frequency Band	Dosimetric Quantity	Limit	Averaging Constraints
100 kHz – 300 GHz	Whole Body SAR	0.08 W/kg	30 min
100 kHz – 6 GHz	Local SAR (Head/Torso)	2.0 W/kg	6 min, 10g mass
100 kHz – 6 GHz	Local SAR (Limbs)	4.0 W/kg	6 min, 10g mass
> 6 GHz – 300 GHz	Absorbed Power Density (S_{ab})	20 W/m ²	6 min, 4 cm ²
> 30 GHz – 300 GHz	Focal Power Density	40 W/m ²	6 min, 1 cm ² (Beam restriction)

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8. Conclusion: The Convergence of Physics and Biology

The study of frequency-dependent human body interactions is a testament to the intricate

coupling between the inanimate laws of Maxwell and the animate complexity of physiology. From the deep, volume-filling induction of power line frequencies (mediated by the α -dispersion of counter-ions) to the superficial, intense absorption of millimeter waves (mediated by the γ -dispersion of water rotation), the frequency of the wave dictates the biological destiny of the energy.

The work of Gabriel et al. provided the necessary map for this terrain, quantifying the dielectric landscape that allows modern dosimetry to function. Yet, the field remains dynamic. The tension between the ICNIRP thermal-based regulatory framework and the growing literature on ion channel dysfunction and oxidative stress represents the central scientific dialectic of bio-electromagnetics today.

As we move toward 6G and THz technologies, where the skin becomes the sole interface and wavelengths approach the scale of cellular structures, the models will need to evolve. The SAM phantom may give way to virtual anatomical twins; SAR may be fully supplanted by thermal modeling; and the "non-thermal" effects may eventually be integrated into a new, more holistic understanding of biological susceptibility. Until then, rigorous adherence to dosimetric precision and continuous surveillance of the biological literature remains the only viable path forward.

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